

Assignment :- 2

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Semester: IV

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Subject: Operating System

Q1) FCFS Scheduling

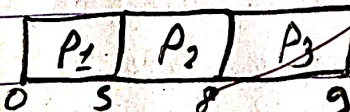
Process	A.T	B.T
P ₁	0	5
P ₂	2	3
P ₃	4	1

And the

- Gantt chart
- waiting time
- turn around time
- Average w.T.

Solution,

			(C.T - A.T)	(T.T - B.T)	
<u>Caution,</u>					
A.	A.T	B.T	C.T	T.T	W.T
P ₁	0	5	5	5	0
P ₂	2	3	8	6	3
P ₃	4	1	9	5	4

Gantt chart : 

completion time: $P_1 = 0 + 5 = 5$
 $P_2 = 5 + 3 = 8$
 $P_3 = 8 + 1 = 9$

Average (w.T) = $\frac{0 + 3 + 4}{3} = 2.33$

Q2

Non-Preemptive SJF

Process	Arrival time	Burst time
P ₁	0	7
P ₂	2	4
P ₃	4	1

Calculate :

- Gantt chart & Average waiting time.

Solution,

(CT-AT) (TT-BT)

P.No	AT	BT	CT	TT	WT
P ₁	0	7	7	7	0
P ₂	2	4	12	10	6
P ₃	4	1	8	8	3

Gantt chart:

0	7	8	12
	P ₁	P ₃	P ₂

Process

$$P_1 = 0 + 7 = 7$$

Process order?

$$P_3 = 7 + 1 = 8$$

P₁ → P₃ → P₂

$$P_2 = 8 + 4 = 12$$

$$\text{Average (WT)} = \frac{0+3+6}{3} = 3$$

③ Preemptive SJF (SRTF)

Process	Arrival time	Burst time
P ₁	0	8
P ₂	1	4
P ₃	2	2

Ans:

- Gantt chart
- Completion Time
- Average Turnaround time

Solution

				(CT-AT)	(TT-B.T)
P.No	AT	BT	CT	TT	W.T
P ₁	0	8	14	14	6
P ₂	1	4	7	6	2
P ₃	2	2	4	2	0

Gantt chart:

P ₁	P ₂	P ₃	P ₂	P ₁
0	1	2	4	7
				14

Process time:

$$P_1: 0+1 = 1 \quad (P_2 = 7 \text{ remaining})$$

$$P_2: 1+1 = 2 \quad (P_2 = 3 \text{ remaining})$$

$$P_3: 2+2 = 4 \quad (\because \text{No other process})$$

$$P_2: 4+3 = 7$$

$$P_1: 7+7 = 14$$

Average turnaround time = $\frac{14+6+2}{3} = 7.33$

④ Priority scheduling (Non-preemptive)

Lower number (∴ 1 = higher priority)

Process	Arrival	Burst	Priority
P ₁	0	5	2
P ₂	1	3	1
P ₃	2	4	3

Ans:

- Gantt chart
- Waiting time

Solution,

P.No	Arrival	B.T	Priority	C.T	(C.T - P.T)	(T.T - B.T)
P ₁	0	5	2	5	5	0
P ₂	1	3	1	8	7	4
P ₃	2	4	3	12	10	6

Gantt chart:

P ₁	P ₂	P ₃
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0 5 8 12

Process: $P_1 = 0 + 5 = 5$ (First run)

$P_2 = 5 + 3 = 8$ (High priority)

$P_3 = 8 + 4 = 12$

∴ Average waiting time = $\frac{0+4+6}{3}$

$$= \frac{10}{3}$$

$$= 3.33$$

low = high priority
number

⑤ Priority scheduling (Preemptive)

Process	Arrival	Burst	Priority
P ₁	0	7	2
P ₂	2	4	1
P ₃	3	1	3

Ans.

• Gantt Chart

• Turnaround time

Solution,

C.T - AT TT - BT

Process	AT	BT	Priority	C.T	T.T	W.T
P ₁	0	7/6.5	2	11	11	4
P ₂	2	4	1	6	4	0
P ₃	3	1	3	12	9	8

Gantt Chart:

P_1	P_1	P_2	P_1	P_3
0	1	2	6	11
				12

Process : P₁ for 0 to 1 = 1

P₁ for 1 to 2 = 2 (∵ P₂ arrives high priority)

P₂ complete = 2 + 4

= 6

(∵ highest priority of all)

P₃ again second highest priority

P₁ = 6 + 5 = 11

P₃ = 11 + 1 = 12

$$\therefore \text{Average (T.T)} = \frac{11 + 4 + 9}{3} = 8$$

⑥ Round Robin (Time Quantum = 2)

Process	Arrival	Burst
P ₁	0	5
P ₂	1	3
P ₃	2	6

(Ans)

Gantt chart, waiting time, average turn around time

Solution

P.No	AT	B.T	C.T	(C.T - A.T)	(T.T - B.T)
				T.T	W.T
P ₁	0	5	12	12	7
P ₂	1	3	9	8	5
P ₃	2	6	14	12	6

Ready Queue:

P ₁	P ₂	P ₃	P ₁	P ₂	P ₃	P ₁	P ₃
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Gantt Chart:

P ₁	P ₂	P ₃	P ₁	P ₂	P ₃	P ₁	P ₃	
0	2	4	6	8	9	11	12	14

Process:

P₁ = 0 + 2 (P₂ and P₃ arrive P₁ still not completed)

P₂ = 1 + 2 (P₂ not completed)

P₃ = 2 + 2 (P₃ not completed)

P₁ = 4 + 2 (P₁ not completed)

P₂ = 6 + 1 (P₂ completed)

P₃ = 8 + 2 (P₃ still not completed)

P₁ = 10 + 2 (P₁ completed)

P₃ = 12 + 2 (P₃ completed)

Average turn around

time = $\frac{12 + 8 + 12}{3}$

= 10.67

⑦ Round Robin (Time Quantum = 3)

Process Burst Time

P₁ 10

P₂ 5

P₃ 8

All arrive at Time 0

calculate:

- Gantt Chart

- Average waiting time

(CT - AT) (TT - BT)

Solution

P.NO. A.T B.T C.T T.T W.T

P₁ 0 10 7 1 23 23 13

P₂ 0 5 2 0 14 14 9

P₃ 0 8 3 2 22 22 14

Ready chart: [P₁ | P₂ | P₃ | P₁ | P₂ | P₃ | P₁ | P₃ | P₁]

Gantt chart: [P₁ | P₂ | P₃ | P₁ | P₂ | P₃ | P₁ | P₃ | A]
0 3 6 9 12 14 17 20 22 23

Process:

P₁ = 0 + 3 = 3 (P₁ remaining (P₂, P₃ arrived))

P₂ = 3 + 3 = 6 (P₂ remaining)

P₃ = 6 + 3 = 9 (P₃ remaining) Average waiting

P₁ = 9 + 3 = 12 (P₁ remaining) Time = 13 + 9 + 14

P₂ = 12 + 2 = 14 (P₂ completed) 3

P₃ = 14 + 3 = 17 (P₃ remaining) (17 - 8) = 9

P₁ = 17 + 3 = 20 (P₁ remaining) (20 - 10) = 10

P₃ = 20 + 2 = 22 (P₃ completed) 8 + 0 = 8

P₁ = 22 + 1 = 23 (P₁ completed) (23 - 10) = 13

⑧ Mixed Arrival (SJF preemptive)

Process Arrival Burst

P₁ 0 6

P₂ 1 3

P₃ 2 7

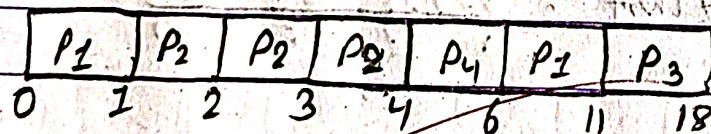
P₄ 3 2

Find: completion time & waiting time?

Solution,

P.N.O	A.T	B.T	C.T	T.T	W.T
P ₁	0	6.5	11	11	5
P ₂	1	2.25	4	3	0
P ₃	2	7	18	16	9
P ₄	3	2	6	3	1

Gantt chart:



Process:

0-1 → P₁

1-4 → P₂ (shortest)

4-6 → P₄

6-11 → P₁

11-18 → P₃

$$\text{Average (T.T)} = \frac{11 + 3 + 16 + 3}{4}$$

$$= 8.25$$

$$\text{Average (W.T)} = \frac{5 + 0 + 9 + 1}{4}$$

$$= 3.75$$

⑨ FCFS with Idle time

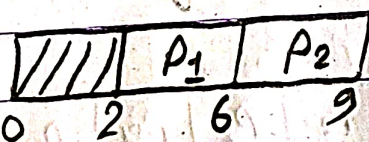
process	arrival	Burst
P ₁	2	4
P ₂	5	3

Ans.

- Gantt chart (include idle time)
- turnaround time

Solution

P. NO	arrival	Burst	C.T	T.T	W.T
P ₁	2	4	6	4	0
P ₂	5	3	9	4	1

Gantt chart : 

Process

CP idle from 0 → 2

2 → 6 → P₁

6 → 9 → P₂ completed

(10) Compare Algorithms

Given:

Process	Burst time
P ₁	6
P ₂	8
P ₃	7
P ₄	3

All arrive at time = 0

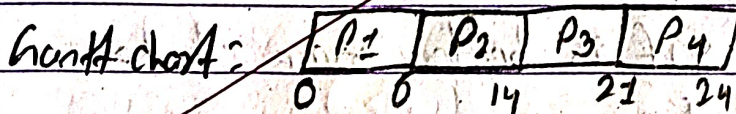
compare: FCFS VS SJF

Find: • Average waiting time for both
• which is better?

Solution,

for
FCFS

P.NO	A.T	B.T	C.T	T.T	W.T
P ₁	0	6	6	6	0
P ₂	0	8	14	14	6
P ₃	0	7	21	21	14
P ₄	0	3	24	24	21



Process order: P₁ → P₂ → P₃ → P₄

$$\begin{aligned} \text{Average waiting time for FCFS} &= \frac{0 + 6 + 14 + 21}{4} \\ &= 10.25 \end{aligned}$$

for
SJF

P.No	AT	B.T	C.T	RT	WT
P ₁	0	6	9	9	3
P ₂	0	8	24	24	16
P ₃	0	7	16	16	9
P ₄	0	3	3	3	0

Gantt chart:

P ₄	P ₁	P ₃	P ₂
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 3 9 16 24

Process order: P₄ → P₁ → P₃ → P₂

$$\therefore \text{Average waiting time (SJF)} = \frac{0 + 3 + 9 + 16}{4} = 7$$

Hence,

The waiting time for FCFS = 10.25 and
for SJF is = 7

That's why SJF is better since it has
less waiting time.

Ans
3.2026
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